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THE FAILURE OF BRITTLE SOLIDS CONTAINING SMALL CRACKS UNDER COMPRESSIVE STRESS STATES

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Abstract—Brittle materials (ceramics, rocks and ice are examples) may contain a distribution of small, grain-sized, cracks. When loaded in compression, these cracks propagate stably until they interact to give final failure. A model is developed for the growth and interaction of cracks in brittle solids under compressive stress states. A critical stress is required to initiate crack growth: it depends on the initial crack length and orientation, on the coefficient of friction and on the stress state. The cracks then grow in a stable way until they start to interact; interaction increases the stress intensity driving crack growth and leads to instability and final failure. This chain of events is modelled, and the framework of a theory of damage mechanics is suggested.

Résumé—Les matériaux fragiles (céramiques, roches et glace par exemple) peuvent contenir une répartition de petites fissures de la taille des grains. Lorsqu'on les soumet à une compression, ces fissures peuvent se propager de manière stable jusqu'à ce qu'elles interagissent pour produire la rupture finale. Nous développons un modèle pour la croissance et l'interaction des fissures dans des solides fragiles soumis à une contrainte de compression. Il faut atteindre une contrainte critique pour amorcer la croissance des fissures: elle dépend de la longueur initiale de la fissure et de son orientation, du coefficient de frottement et de l'état de la contrainte. Les fissures croissent alors de maniere stable jusqu'à ce qu'elles commencent à interagir; cette interaction augmente l'intensité de la contrainte entraînant la croissance des fissures et conduit à une instabilité et à la rupture finale. Nous modélisons cette suite d'événements et nous esquissons une théorie de la mécanique de l'endommagement.

Zusammenfassung—Spröde Materialien (Keramik, Gestein und Eis sind Beispiele) können kleine Risse in der Größe der Körner in unterschiedlicher Verteilung enthalten. Werden diese Materialien im Druck belastet, dann breiten sich die Risse stabil aus, bis sie miteinander wechselwirken und zum Bruch der Proben führen. Für das Wachstum und die Wechselwirkung der Risse in spröden Festkörpern unter Druckspannungen wird ein Modell entwickelt. Für den Beginn des Rißwachstums wird eine kritische Spannung benötigt. Diese hängt ab von der ursprünglichen Rißlänge und -orientierung, vom Reibungskoeffizienten und von dem Spannungszustand. Die Risse wachsen dann auf stabile Weise bis sie miteinander wechselwirken. Diese Wechselwirkung erhöht die Spannungsintensität und damit die treibende Kraft des Rißwachstums, welches zu Instabilität und Bruch führt. Diese Ereigniskette wird im Modell beschrieben, wodurch ein Rahmen für eine Theorie der Schädigungsprozesse nahegelegt wird.

NOMENCLATURE

σ_1	largest principal compressive stress (MPa)
σ_2, σ_3	smaller principal compressive stresses (MPa)
σ, τ	normal and shear stress on crack plane, equation (1)
σ_{xy}	shear stress acting in crack plane (MPa)
σ'_{xy}	reduced shear stress, $(\sigma_{xy} + \mu\sigma_{yy})$ (MPa)
σ_{yy}	normal stress acting across crack faces (MPa)
F, F_n	forces tending to open the crack (N)
$2a$	length of initial, angled crack (m)
l	extension of wing crack (m)
L	dimensionless wing crack length $L = l/a$
b	sample thickness (m)
t	width of beam or column, Fig. 17 (m)
ψ	angle between σ_1 and crack face (Fig. 11)
θ	angle measured from crack plane (Fig. 11)
E_0	Young's modulus of uncracked body (MPa)
E	Young's modulus of the cracked body (MPa)
ν	Poisson's ratio
K_I	mode I stress intensity (MPa $m^{1/2}$)
K_{IC}	mode I fracture toughness (MPa $m^{1/2}$)
u_x	crack sliding displacement (m)
$2\delta_n$	mode I crack opening displacement at mid point (m)
$2\delta_s$	mode II sliding displacement at mid point (m)
$2\delta_T$	net mode I crack opening displacement at mid point (m)
T_n, T_s	tractions acting on the crack surfaces (MPa)
G	strain energy release rate (J m^{-2})
μ	coefficient of friction
U, U_I, U_{II}	elastic energy stored in crack fields (J)
W	work done by applied stresses (J)
N_A	number of cracks per unit area (m^{-2})
λ	ratio of principal stresses, σ_3/σ_1
σ_y	yield strength (MPa)
$\alpha_1, \alpha_2, \alpha$	dimensionless constants
β_b, β_s	dimensionless constants

1. INTRODUCTION

Cracks, holes and inclusions in an elastic solid can interact with a compressive stress field in a way which causes new cracks to grow from them. If these cracks extend to the sample surface, or if they interact with each other so that they grow unstably, then a macroscopic failure may follow. The topic is one which has

attracted research in many fields: materials science and mechanics [1, 2], geology [3–7] and the branches of engineering which relate to the use of cement and concrete [8, 9].

This paper is concerned with crack growth, in compression, from pre-existing cracks. (The growth of cracks in compression from pores and holes is analysed in a companion paper [10]). Crystalline ceramics, rocks and minerals often contain a distribution of fine cracks with a size about equal to the grain size: they are caused by thermal or elastic stress during earlier thermal or mechanical loading. Acoustic and dilatometric studies show that they start to propagate when the axial compressive stress reaches about one half of the ultimate failure stress, and microscopic observations show that they extend parallel to the compression-axis (see, for example, Paterson [11]).

Their subsequent behaviour depends on the confining pressure, as shown in Fig. 1. Simple axial or radial compression, shown at (a) and (d), cause a few cracks to propagate and combine to give failure on planes parallel to the maximum compressive stress ("slabbing"). A modest confining pressure prevents this unlimited crack growth; failure then occurs by the interaction of cracks to give the macroscopic shear failure shown at (b). Larger confining pressures limit the growth of individual cracks even further, and the sample deforms in a pseudo-ductile way with large-scale deformation taken up by many short, homogeneously distributed microcracks, shown at (c).

The tensile fracture of a brittle solid is relatively simple and well understood: a single flaw (the longest or most favourably oriented) propagates unstably when the stress intensity, K_I, at its tip exceeds the critical stress intensity (or *fracture toughness*) K_{IC} of the solid. The mechanics of compressive brittle fracture is more complex and less well understood. Many attempts to model it start from the stress field of an elliptical flaw, and calculate the conditions under which a crack first initiated at some point on its surface [1, 4–6, 12]: this initiation stress was assumed to correspond to failure. But the propagation of axial microcracks is, as we shall see below, a stable process [2, 8, 13] and they propagate in such a way that they do not readily link [13]. The co-operative process by which these microfractures produce macroscopic failure has remained one of the more elusive problems of modern fracture mechanics: but it is known that final failure occurs at a stress which can be much larger than that at which the cracks first start to grow, so that a proper treatment of failure must include consideration of crack initiation, crack propagation and crack linkage.

In this paper we analyse the growth of a crack from an initial inclined flaw and examine the conditions under which an array of such cracks interact. The method is an approximate one, using well-known results of fracture mechanics and beam theory, combined in simple models. It is less accurate than numerical methods based on finite-element computations or on the integration of appropriate Green's functions over the crack surfaces (as used in the important paper by Nemat-Nasser and Horii [2], with which we compare our results). Its merit lies in its simplicity and in the way it leads to a simple treatment of the interaction of cracks with surfaces and with each other; and the results agree, with a precision which is certainly greater than that of the experimental data, with the more accurate calculations. The theoretical results are compared with experiments using brittle plastics, which in turn suggest extensions to the models.

It should be emphasised that compression-failure of brittle solids is a very complicated process. This

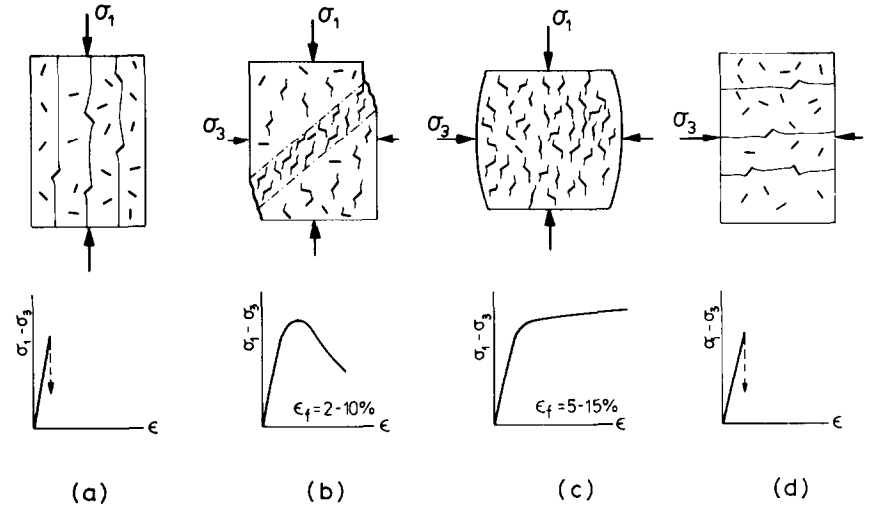


Fig. 1. Failure modes in compression. (a) and (d) Slabbing, when one or a few cracks propagate parallel to the principal compression, (b) failure by aggregation of cracks to form a shear zone and (c) near-homogeneous deformation by distribution microcracking.

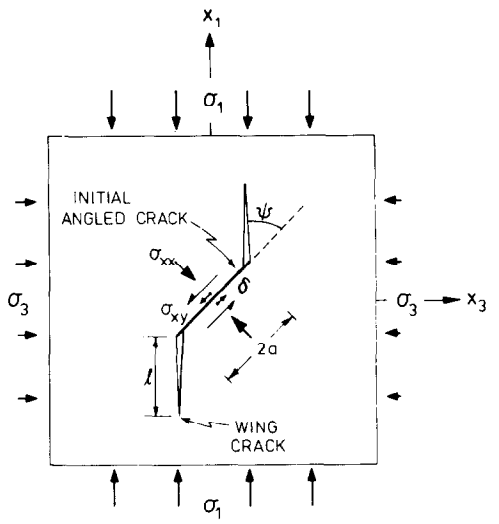


Fig. 2. Schematic of wing crack growth from an angled crack under a compressive stress σ_1.

paper and the companion [10] leave many aspects of the problem unexplained, but the approach appears to have much promise, and to be capable of further development.

2. EXPERIMENTS: CRACKING FROM AN ANGLED CRACK IN A PLATE IN COMPRESSION

We studied the growth of wing cracks from sharp starter cracks in PMMA plates, shown schematically in Fig. 2. Figure 3 shows photographs of the way the wing cracks extend stably from the initial, angled crack as the compressive load (applied vertically) was increased.

The plates were 10 mm thick, 170 mm high and 25 to 100 mm wide. Cracks were made by drilling a 1.5 mm hole and extending either side to 14 ± 0.5 mm with a fret saw. The crack was extended further by about 1 mm at each end by tapping a razor blade placed in the slot. Two brass shims were inserted into the slot to control friction and to ensure contact of the two surfaces (essential when simulating the behaviour of a closed crack). Samples were prepared with single 16 mm starter cracks at angles ψ of 15, 30, 36, 45, 60 and 75° to the compression axis, and with various arrays of starter cracks.

The plates were loaded in axial compression in a rigid jig with grips to prevent out-of-plane displacement or rotation of the ends of the sample, and with edge-guides to suppress buckling [13]. The behaviour of the growing crack is extremely sensitive to the end conditions. Reliable results were obtained by loading through slotted rubber pads, and by lining the guiding faces with teflon. Crack growth was photographed and correlated with the load–deflection curve.

Typical data for single-crack growth are plotted in Figs 4 and 5, using normalised axes. The length of the wing cracks is plotted in units of a, the half-length of the initial crack; and the compressive stress σ_1, is plotted in units of $K_{IC}/\sqrt{\pi a}$ where K_{IC} is the fracture toughness of the PMMA ($K_{IC} = 1.0$ MPa m$^{1/2}$ for slow crack growth [14]). Figure 4 shows the influence of the angle ψ of the starter crack: cracks which lie near 45° to the compression axis grow most easily, though all those in the range $30 < \psi < 60$ nucleated wings at nearly the same stress. Figure 5 shows the influence of the width of the sample: the cracks grow much more easily in narrow samples, indicating a strong interaction between the crack and the surface parallel to which it grows (an observation which is critical in understanding the interaction between cracks).

Figure 6 is typical of the observations of crack growth in an array of starter cracks. The data are

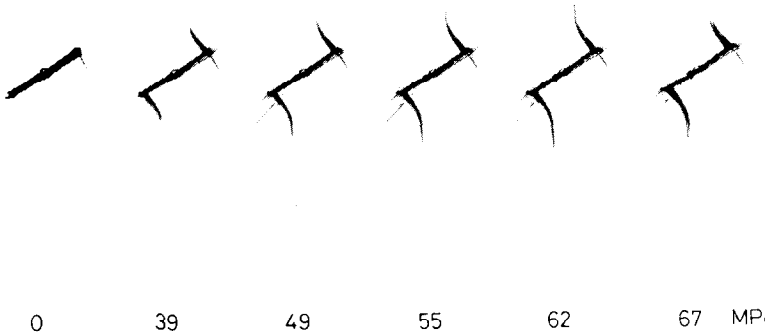


Fig. 3. A sequence of photographs of wing crack growth under progressively higher compressive stress.

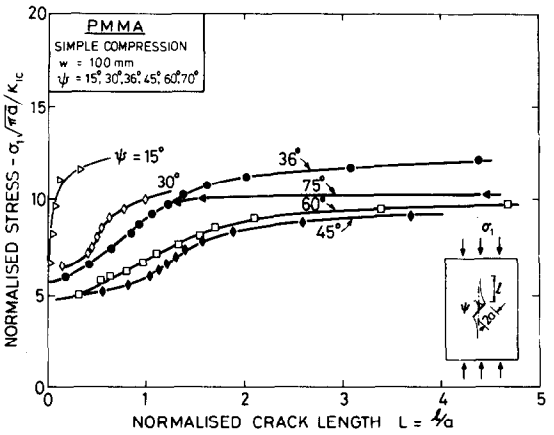


Fig. 4. Wing crack growth in PMMA in simple compression, showing the influence of the initial crack angle, ψ.

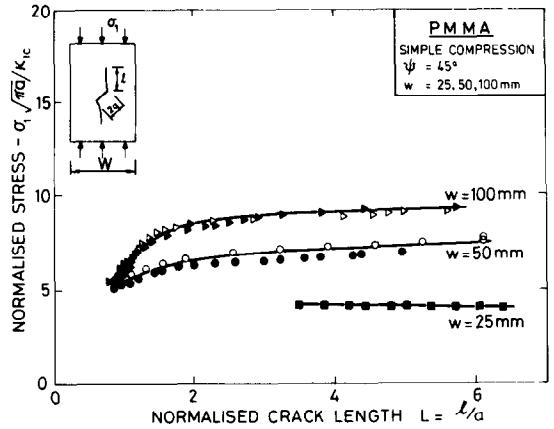


Fig. 5. Wing crack growth in PMMA in simple compression, showing the influence of the specimen width, w.

plotted in Fig. 7. In all cases, the wing cracks at first extend rapidly at a load well below that for an isolated crack, but stopped when their tips were level with the centre of the neighbouring cracks. Increasing the loads from this point produces only small increase in crack length.

Three further observations are relevant. First, the reproducibility of the data is good. Figure 8 shows the results of three nominally identical runs. The stress, for a given crack length, is reproducible to within $\pm$ 6%. Second, when narrow specimens are loaded, there is an outward bending of the ligaments on either side of the growing wing cracks. This observation, shown in Fig. 9, is important in understanding the interaction of cracks with surfaces and with other cracks (analysed later). And third, PMMA is not an ideally brittle solid; its stress–strain curve is shown in Fig. 10, measured by compressing uncracked samples at the same loading rate as that used for the other tests (equivalent to a strain rate of 2×10^{-5}/s). The material is linear-elastic to 60 MPa, but then deforms non-linearly to a maximum (with full plasticity) at 94 MPa.

3. ANALYSIS: INITIATION AND GROWTH OF WING CRACKS IN A PLATE IN COMPRESSION

This section summarises the analysis of the growth of wing cracks from an initial flaw in a plate subjected to a remote principal stress (σ_1, σ_3), as shown in Fig. 2. The crack lies at an angle ψ to the most compressive principal stress, σ_1. The initiation and the growth of wing cracks in an infinite plate are considered first; then the interaction of the growing wing cracks with the edges of a finite plate is analysed; and finally the influence of plasticity is discussed. The physical basis and the results of the analysis are given here. The mathematical details can be found in the Appendix.

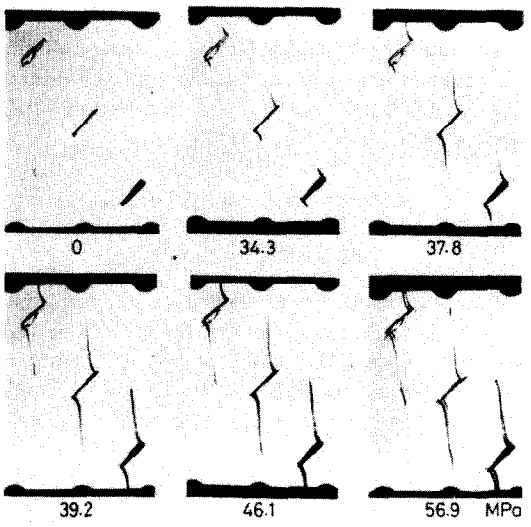


Fig. 6. Wing crack growth from an array of initial cracks in PMMA.

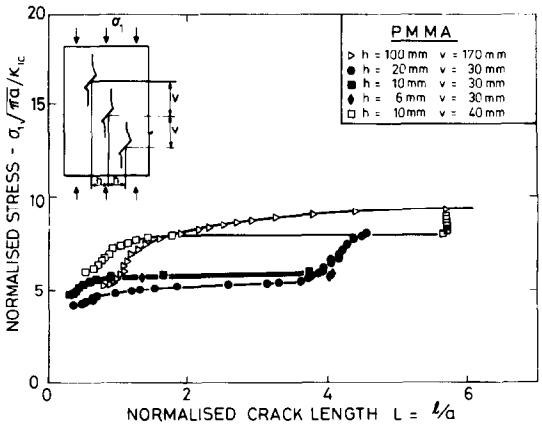


Fig. 7. Typical data for wing crack growth from an array of initial, angled, cracks. The wing cracks grow more easily when another crack is nearby.

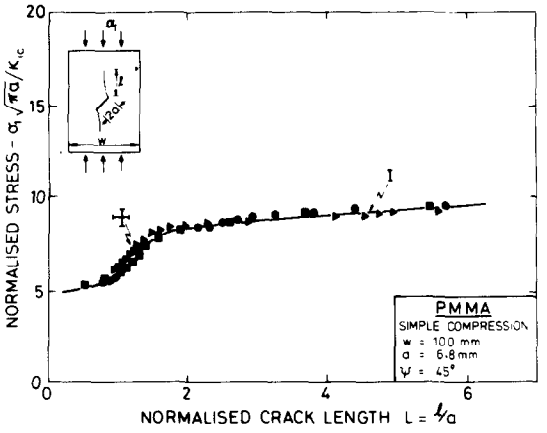


Fig. 8. The reproducibility of the data. The experimental scatter is greatest when the wing cracks are very short, but even here the scatter is much less than the differences introduced by changing ψ or w (Figs 5 and 6).

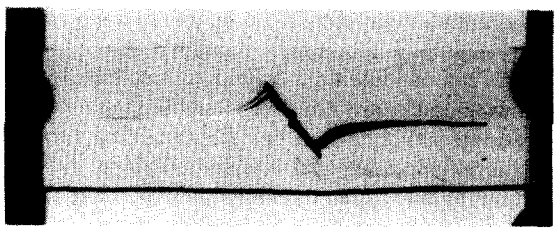


Fig. 9. The outward bending of the unconstrained sides of the sample when l/w becomes greater than unity.

3.1. *Initiation of wing cracking in an infinite plate*

Consider an infinite elastic plate containing an initial crack of length $2a$ and subjected to principal stresses σ_1 and σ_3. The stresses are treated as positive when tensile, negative when compressive; then σ_1 is the most negative (most compressive) and σ_3 the most positive (least compressive) principal stress. The crack lies at an angle ψ to the X_1 axis, and contains the X_3 direction as shown in Fig. 11. The remote stress field generates shear (σ_{xy}) and normal (σ_{xx}) stresses on the crack plane, where

$$\sigma_{xy} = \frac{\sigma_3 - \sigma_1}{2} \sin 2\psi = \tau \sin 2\psi \qquad (1a)$$

and

$$\sigma_{xx} = \frac{\sigma_3 + \sigma_1}{2} + \frac{\sigma_3 - \sigma_1}{2} \cos 2\psi = \sigma + \tau \cos 2\psi. \qquad (1b)$$

(It is implicit in the sign convention that σ_{xy} is always positive and σ_{xx} is always negative.) The shear stress tends to make the crack surfaces slide; but because the cracks are closed, a frictional stress $\mu\,\sigma_{xx}$ opposes the sliding (μ is the coefficient of friction). Then the effective sliding stress is

$$\sigma'_{xy} = \sigma_{xy} + \mu\,\sigma_{xx}. \qquad (2)$$

This stress is intensified by the crack, so that a singular stress field, characterised by the quantity $\sigma'_{xy}\sqrt{\pi a}$, appears at its tips. In the plane of the initial crack, this field is predominantly shear in character; but on planes which lie at an angle θ to the crack tip (Fig. 11) a normal stress appears, tending to cause a mode I wing crack to grow from the tips of the initial, angled flaw.

The stress field at the tip of a sliding crack is known [15]. The mode I stress intensity, K_I, is found by seeking the plane, θ, on which the tensile stress is a maximum; and the most critical flaw is then identified by maximising again, this time, with respect to the flaw angle, ψ, (Fig. 11). This is done in the Appendix, giving the maximum K_I as

$$K_I = -\frac{\sigma_1\sqrt{\pi a}}{\sqrt{3}}\{(1-\lambda)(1+\mu^2)^{1/2} - (1+\lambda)\mu\} \qquad (3)$$

on the plane $\theta = 70.5°$ for the crack of angle $\psi = \frac{1}{2}\tan^{-1}(1/\mu)$. Here λ is the ratio of the principal stresses: $\lambda = \sigma_3/\sigma_1$. The condition for crack initiation is given by setting K_I equal to K_{IC}, (the fracture toughness of the material) giving the *initiation condition*

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{-\sqrt{3}}{[(1-\lambda)(1+\mu^2)^{1/2} - (1+\lambda)\mu]} \qquad (4)$$

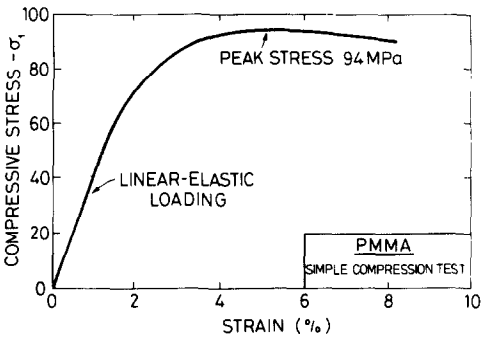


Fig. 10. A stress-strain curve for uncracked PMMA in compression.

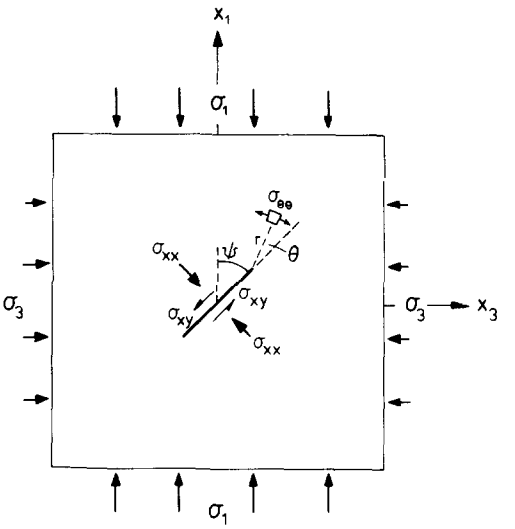


Fig. 11. The co-ordinates, stresses and angles involved in analysing wing-crack initiation.

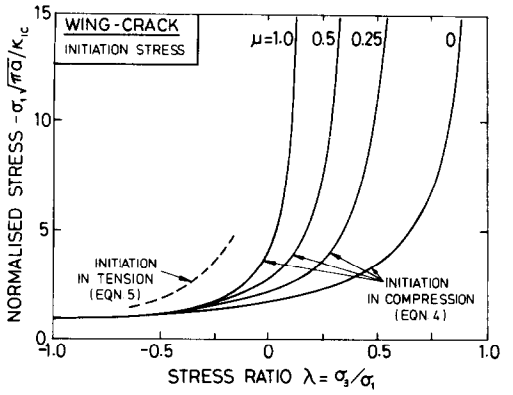


Fig. 12. The condition for the nucleation of wing cracks, for various values of the coefficient of friction [equation (4)]. The tensile cut-off (broken line) intersects the compression-nucleation lines (full lines) at $\lambda = -1$.

when σ_3 is sufficiently positive, a tensile fracture will occur from cracks where $\theta = \psi = 0$ and the initiation condition becomes

$$\sigma_3\sqrt{\pi a} = K_{IC}$$

or

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = -\frac{1}{\lambda}. \tag{5}$$

This second criterion truncates the first [equation (4)].

These results are identical with that of Nemat-Nasser and Horii (1982), and similar in form to those of McClintock and Walsh (1962) (the constant $\sqrt{3}$ of our equation (4) is replaced by a 4 in theirs because they use a limiting solution for the stress field at an elliptical flaw which breaks down when the crack is sharp). They are plotted as a function of λ and μ in Fig. 12.

Strictly, equation (4) is only valid when σ_{xx} is compressive. When it is tensile, the crack opens and the frictional force $\mu\sigma_{xx}$ disappears: in addition, the normal stress σ_{xx} is now concentrated by the crack giving a new term in the equation for the stress intensity. This calculation is covered in detail in the Appendix and actually provides a smooth transition from the closed crack solution to the tensile truncation. For most purposes, it is adequate to approximate crack initiation by equation (4) truncated by equation (5).

3.2. *Growth of wing cracks in an infinite plate*

Once wing cracks have initiated, further sliding of the main crack wedges them open, causing them to grow further. The extent of growth, l, is calculated by the method illustrated in Fig. 13. When the sample is loaded, the crack slides, allowing the remote stresses σ_1, σ_3 to do work. The sliding generates an internal stress field in which elastic energy is stored. The equilibrium sliding displacement ($\pm\delta$) is calculated by minimising the potential energy of the body (the work done minus the energy stored) with respect to δ. Part of the displacement δ can be thought of as a wedge-opening, applied to the mouth of each wing crack. Standard results of fracture mechanics can then be used to calculate K_I at the tip of a wing crack. The wing crack grows until K_I falls to K_{IC}.

There is one additional complication. Even when the wing-cracks are vanishingly small, there exists a stress intensity K_I at their tips: it is caused by the field of the initial crack, and was calculated in the last section. It dominates the early stage of wing crack growth. The final expression for K_I therefore has two contributions: one from the stress field of the initial, angled crack, dominant when l is small; the other from the wedging action caused by the sliding displacement δ, dominant where l is large.

The analysis is given in the Appendix. The important result is the *stress intensity driving wing crack growth in an infinite plate*, it is

$$K_I = \frac{-\sigma_1\sqrt{\pi a}}{(1+L)^{3/2}}\{1-\lambda-\mu(1+\lambda)-4.3\,\lambda L\} \times\left\{0.23\,L+\frac{1}{\sqrt{3}(1+L)^{1/2}}\right\}. \tag{6}$$

The wing cracks grow until K_I falls to K_{IC}, giving the

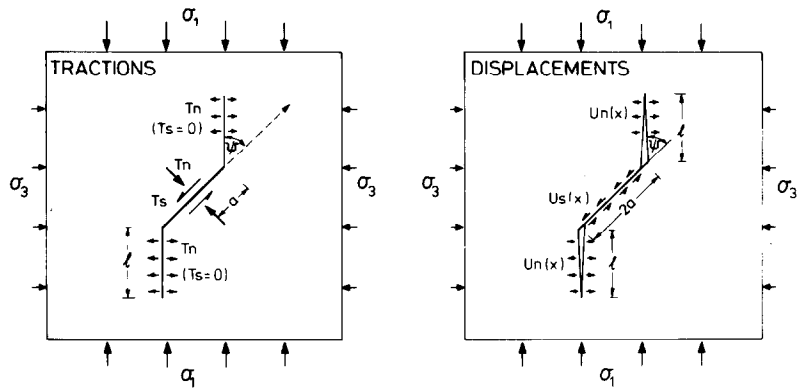


Fig. 13. Wing crack growth, showing the tractions (T_n and T_s), the displacements U_n and U_s. The maximum sliding displacement at the centre of the crack is $\pm\delta$.

relationship between axial stress and crack length

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{-(1+L)^{3/2}}{[\{1-\lambda-\mu(1+\lambda)-4.3\,\lambda L\} \times \{0.23\,L + 1/(\sqrt{3}(1+L)^{1/2})\}]}. \quad (7)$$

Physically, this result incorporates three regimes: when L is zero, K_I is determined by shear of the initial crack; then equation (7) reduces to the initiation criterion, equation (4). When L is about unity (though it depends on λ), K_I is dominated by the wedging open of the wing cracks caused by the sliding of the initial angled crack, and is not much influenced by the confining stress, σ_3. Finally, when L is large, K_I is greatly influenced by σ_3 [because it acts on the entire crack length $2(l+a)$, whereas σ_1 can be thought of as acting only on the angled segment of the crack, of length $2a$]. When σ_3 is zero or compressive, crack growth is always stable; but when σ_3 is tensile (even when it is very small), the crack eventually becomes unstable. When λ is zero (simple compression) the crack grows stably such that, for large L

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{-\sqrt{L}}{0.23(1-\mu)}.$$

Equation (7) is plotted as full lines in Figs 14, 15 and 16, for three values of the coefficient of friction, μ. Where possible, our results are compared with those of Nemat-Nasser and Horii [2]† who used a Green's-function method (broken lines). Figure 14 is for zero friction. It shows a plot of normalised stress $-\sigma_1\sqrt{\pi a}/K_I$ against $L = l/a$ for nine values of λ, on log scales. At $L = 0$, this result and the result of Nemat-Nasser and Horii are identical; and they are identical, too, when $L \gg 1$. The figure shows that the approximate calculation is in good agreement throughout the entire range of interest. Figures 15 and 16 show similar plots and comparisons for $\mu = 0.3$ and $\mu = 0.6$. It is clear that the physical model underlying the approximate equation properly describes the important features of the dependence of $-\sigma_1\sqrt{\pi a}/K_I$ on L. Figure 14 also shows the results of the approximate analysis proposed by Steif [16] (dotted line). It has roughly the same shape as our equation but differs from it, and from Nemat-Nasser and Horii's results, by a factor of two or more.

The important features of the result are immediately obvious from these figures. Crack growth is stable if λ is zero or positive, but even a very small lateral tension (λ negative) leads to a peak beyond which crack growth becomes unstable.

3.3. *Wing crack growth in a finite plate: interaction with surfaces*

The experiments of Section 2 are compared with the predictions of the theory a little further on. The comparison shows that the wing cracks grow in the way described by equation (7) (and plotted in Figs 14–16) when they are short, but that they grow faster than this when their length l becomes comparable with the width, w, of the plate. We have studied the behaviour of angled cracks with wings in silicone rubber plates (in which large elastic deflections are possible) and made detailed measurements of the shape-change of the PMMA plates during loading. The observations and measurements all indicate that crack growth accelerates when additional *bending displacements* appear in the plate. These bending displacements can be seen in Fig. 9, and are shown schematically in Fig. 17. We now analyse this crack-surface interaction and show how it explains the experimental observations.

Consider a plate containing a central angled crack with principal stresses σ_1, σ_3, as shown in Fig. 17(a). If the dimensions, h, w of the plate are both much larger than the length $2a$ of the wing crack, the stress intensity at its tip is given by equation (6). But if the

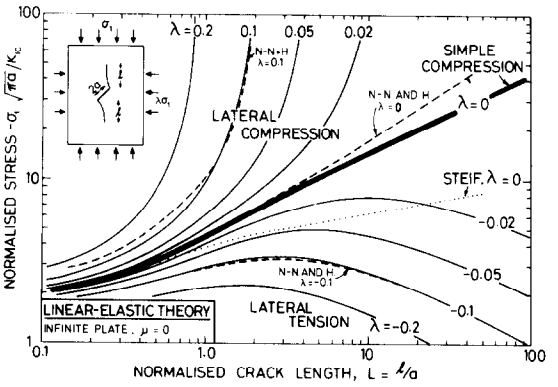


Fig. 14. Wing crack growth in an infinite plate [equation (7)] with $\mu = 0$. The approximate equation developed in the text (full lines) is in good agreement with the numerical calculations of Nemat-Nasser and Horii [2] (broken lines). Steif's [16] approximate equation (dotted line) has the same shape but lies at much lower stresses.

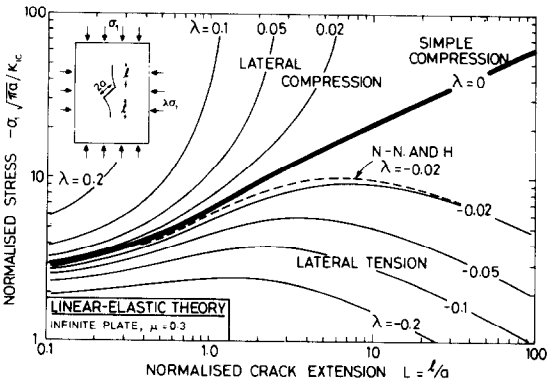


Fig. 15. Wing crack growth in an infinite plate [equation (7)] with $\mu = 0.3$. The approximate equation agrees well with the numerical calculation of Nemat-Nasser and Horii [2].

†This comparison is with corrected computations, kindly provided by Professor Nemat-Nasser. The original Fig. 9 of Nemat-Nasser and Horii [2] is incorrect.

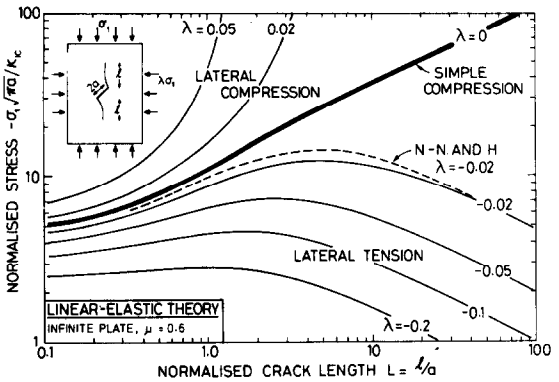


Fig. 16. Wing crack growth in an infinite plate [equation (7)] with $\mu = 0.6$. Again, agreement with the numerical results of Nemat-Nasser and Horii [2] is good.

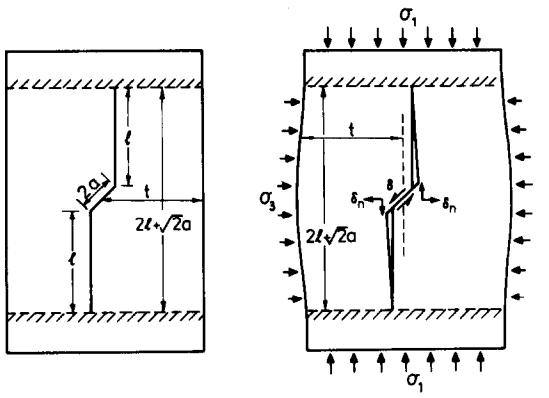


Fig. 17. When the sample is narrow, the wing cracks split into vertical beams or columns, built-in at either end. Sliding at the initial crack bends the beams; the bending displacements increase the stress intensity K_I.

plate is narrow, the wing cracks split it into vertical beams or columns of width $t \simeq w/2$ and length $2(l + \alpha a)$ where $\alpha = \cos\psi$. On loading the plate, sliding at the initial crack wedges open the wing cracks as before. But the beams, being narrow, bend outwards—the narrower they are, the more easily they bend—allowing a larger sliding displacement than was possible in the infinite plate. This extra displacement increases the stress intensity at the crack tip and makes it grow further (at the same load) than before.

We calculate the extra contribution to K_I as follows. If the sample is first loaded with the crack faces pinned together, and then the crack faces are released so that they slide and separate, the potential energy of the body and its loads changes. If this potential energy is minimised with respect to sliding displacement, δ, the stable value of δ can be found. This is done in the Appendix† by treating each half of the sample in Fig. 17(a) as a beam which is bent and is compressed by the remote stresses σ_1, σ_3. The stable value of δ is then used to evaluate the elastic energy in one beam, U, and the crack driving force G is calculated from

$$G = -\frac{1}{b}\frac{dU}{dl}.$$

Finally, the contribution to K_I is calculated from $K_I = \sqrt{EG}$, with the result

$$K_I^B = -\frac{1}{2\sqrt{2\pi}}\left(\frac{a}{t}\right)^{1/2}\sigma_1\sqrt{\pi a} \times \left\{\frac{1 - \dfrac{4}{(3\sqrt{2})^{1/2}}\dfrac{a}{t}(L + 1/\sqrt{2})^2\lambda}{1 + \dfrac{12}{\pi^2}\dfrac{a^2}{t^2}(L + 1/\sqrt{2})^2\dfrac{\sigma_1}{E}}\right\} \quad (8)$$

where $t = w/2$ and $L = l/a$. This expression describes the interaction of the crack with the surfaces of the sample, in a multiaxial stress field, including elastic collapse. It consists of three parts. The part in front of the curly brackets is the contribution to K_I from the outward bending caused by σ_1. The term above the horizontal line in the curly brackets describes the restraining effect of the confining stress σ_3. Finally, the term below the horizontal line is a factor which corrects for the change in geometry caused by beam bending: it goes to zero (causing K_I^B to go to infinity) at the Euler buckling load of the column. Note that K_I^B is negligible for wide plates ($t/a \gg 1$), but becomes increasingly important as the plate width is reduced.

The total K_I driving crack growth is given by adding equation (8) to (6). As before the crack grows until K_I falls to K_{IC}.

This completes the elastic theory for an isolated crack. But, as shown in Section 2, PMMA can become plastic if the compressive stress is large enough. When a beam of thickness t and depth b is subjected to an axial stress σ_a and a bending moment m, it starts to yield when the maximum surface stress reaches the yield strength, that is, when

$$\sigma_a + \frac{6m}{bt^2} = \sigma_y.$$

The beams shown in Fig. 17 are narrowest at the initial angled crack. Here the axial stress is

$$\sigma_a = \frac{\sigma_1}{(1 - \alpha a/t)} \approx \sigma_1$$

and the bending moment, locally, at the narrowest part of the beam, is

$$m = \frac{\sigma_1 bt\,\alpha a}{2}$$

giving the condition for the onset of plasticity at the narrow section of the beam as

$$\sigma_1 = \frac{\sigma_y}{1 + 3\alpha\, a/t} \quad (9)$$

†For simplicity we take the most damaging crack to be that of an angle $\psi = 45$, as suggested by the experimental data plotted in Fig. 4.

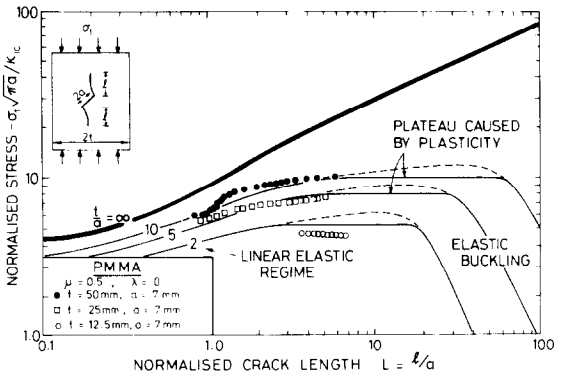


Fig. 18. Comparison of theory with experiments for PMMA.

where $\alpha = 1/\sqrt{2}$. When yielding starts, the crack is wedged open further, and K_I increases greatly. This is shown on subsequent figures which describe the PMMA data by truncating the elastic solution at the stress given by equation (9), that is, at

$$-\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{\sigma_y\sqrt{\pi a}}{K_{IC}(1+3\alpha\, a/t)}. \qquad (10)$$

3.4. Comparison of theory with experiments for cracks in plates

Data for PMMA (Section 2) are compared with theory (full lines) Fig. 18. The linear elastic theory explains the stable crack growth up to a length $L \approx 2$. A truly elastic material (such as glass) would continue to follow the elastic theory (broken line) until buckling caused catastrophic crack growth; the glass samples tested in Ref. [10] behaved in this way. PMMA, however, starts to yield at the beam centres when the yield strength (94 MPa) is locally exceeded, explaining the deviations from the linear elastic theory which give the plateau shown in Fig. 18. The combined elastic–plastic analysis accounts fairly well for the magnitude and shape of the crack-growth data, and explains the dependence of the plateau level on plate thickness, t/a.

4. THE DAMAGE MECHANICS OF CRACKED BRITTLE SOLIDS

Real brittle solids (rocks, cements, concretes, for example) contain a distribution of cracks, pores and inclusions. When loaded in compression, wing-cracks grow from these flaws. At first, when each wing-crack is short, the growth is dominated by the field of the flaw from which it grows. But as it extends it starts to interact with other growing cracks in the population, and this interaction ultimately leads to crack coalescence and final, macroscopic, failure (Fig. 1).

Problems of this sort are usefully approached through the ideas of *damage mechanics*: the mechanics of an array of growing, interacting, cracks. We shall use the method of Sammis and Ashby [10], who develop a framework of damage mechanics to describe the growth of cracks from a distribution of holes in a brittle solid in compression. Briefly, when wing-cracks are short, they grow as if in an infinite body; their growth was analysed in Section 3, leading to equation (7). As they grow longer, they start to interpenetrate, as sketched in Fig. 19, dividing the solid into beams (one is shaded) which become longer and narrower as the cracks extend. If the total number of cracks remains constant, then the area of a beam remains constant also. If the beam length is $(l + a)$, then its width t is given by

$$t(l+a) = \frac{1}{2N_A}$$

where N_A is the number of initial, angled, cracks per unit area. The factor 2 appears because each crack has two beams associated with it.

The centre lines of the beams can be thought of as a network of connected struts (in two dimensions) or of plates (in three), shown in Fig. 19(b). As the cracks grow longer, the ligaments between the cracks are increasingly well approximated by such a network, through which the forces which are applied to the solid are transmitted as a system of axial forces, shear forces and bending moments. The statics and kine-

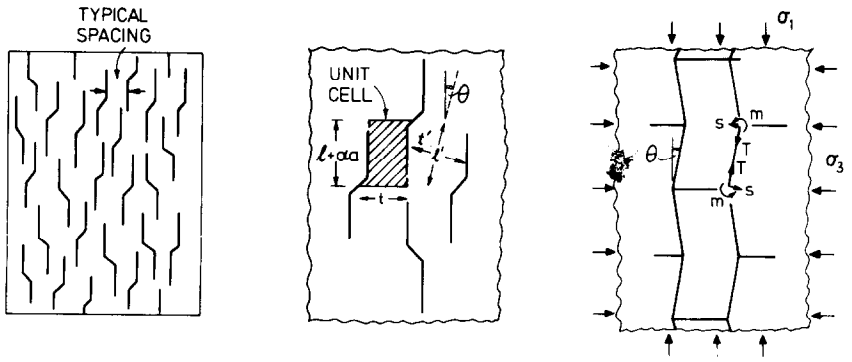


Fig. 19. As cracks in an array extend, the body is divided into ligaments which can be thought of as a network of beams through which the remotely applied stresses are transmitted as a set of axial forces and bending moments.

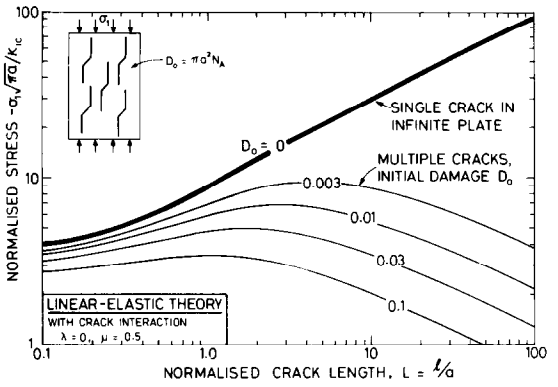


Fig. 20. The stress required to cause cracks to grow to a length L.

matics of the structure can then be analysed in terms of the forces and moments on a single, typical, unit of the network.

Figure 19 shows that the centre-line of the beams zig-zag through the material, making angles of $\pm\theta$ to the vertical (σ_1) direction, where

$$\tan\theta = \frac{\alpha a}{l + \alpha a}$$

and $\alpha = 1/\sqrt{2}$.

When the cracks are short, the unit-strut of the network is short and stocky; it suffers axial compression but no significant bending. Then the cracks do not intereact and the stress intensity is given by equation (3). But as they extend, the unit-beam becomes longer and thinner, and the moments exerted on it by the remote stresses σ_1 and σ_3 cause bending deflections and the cracks interact. This interaction is analysed by Sammis and Ashby [10], using methods which parallel those used in Section 3 of the present paper to analyse the interaction of a crack with a free surface. Applied to the present problem, the method gives the additional contribution to the stress intensity, K_I, caused by crack interaction; it is

$$K_I = -\frac{\sqrt{2}}{\pi} D_0^{1/2}(L+\alpha)^{1/2}\sigma_1\sqrt{\pi a} \times \left\{\left[1 - \frac{8}{\pi} D_0\lambda(L+\alpha)^3\right]\left[1 - \frac{2}{\pi} D_0\lambda(L+\alpha)^3\right]\right\}^{1/2}$$

where D_0 measures the initial level of damage

$$D_0 = \pi a^2 N_A.$$

The term outside the curly brackets is the extra crack-opening stress intensity caused by the major compression σ_1; the term inside describes the crack-closing contribution caused by the confining stress σ_3. The total K_I is the sum of this and equation (3). As before, the crack grows until K_I falls to K_{IC}. The result is plotted in Fig. 20 which shows how the cracks grow stably at first and then interact in a way which makes them unstable.

This begins to explain the observation of crack growth in plates containing many cracks (Figs 6 and 7 and Ref. [13]). There, individual cracks, when short, grow in the same way as isolated cracks (extreme left of Fig. 7, where the data overlap). When longer, the cracks interact in a way which makes their growth easier: the plateaus of Fig. 7 show that extensive growth now occurs at almost constant stress, and that the plateau is low when initial crack density is high. The present theory does not explain the final phase of crack behaviour observed in the experiments: that when one crack has grown until its tip is opposite the mid-point of its neighbour, it becomes stable again, and deviates from the vertical towards the neighbour, with which it then links. This, and the problem of constructing fracture surfaces in stress space will be the subject of a later paper.

5. SUMMARY AND CONCLUSIONS

The growth of brittle cracks from initial crack-like flaws in a compressive stress field has been studied experimentally and analysed by simple approximate methods. As first the cracks grow stably (meaning that the load must be increased to cause increase in crack length) and in a direction parallel to the direction of the largest compressive stress. If one of the other principal stresses is tensile then the crack ultimately becomes unstable and propagates catastrophically.

Experiments show that cracks interact with the surfaces of a finite sample in a way which causes them to grow faster than they do in an infinite medium. A simple analysis based on beam theory is developed to describe this interaction, and is extended (following Sammis and Ashby [10]) to describe crack-crack interaction in a body containing many closely spaced cracks. The method leads to a good description of the experiments on single cracks and suggests the basis for developing a damage mechanics for multiply-cracked brittle solids in compression. These ideas of damage mechanics are developed further in a companion paper [10].

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APPENDIX: CALCULATIONS OF STRESS INTENSITY FACTORS

Wing crack initiation

Consider a crack of length $2a$, contained in an infinite plate and lying at the angle ψ to the larger principal compressive stress σ_1 (Fig. 12). The tensile stress $\sigma_{\theta\theta}$ at a distance r from the tip at an angle θ to the crack plane (15) is

$$\sigma_{\theta\theta} = \frac{3}{2}\frac{\sigma'_{xy}\sqrt{\pi a}}{\sqrt{2\pi r}}\sin\theta\cos\frac{\theta}{2} \qquad \text{(A1)}$$

where σ'_{xy} is the effective shear stress on the crack plane [equation (2)]. Following Cotterell and Rice [17] the mode I stress intensity K_I on a very small wing crack of length l and at an angle θ to the main crack may be approximated by $\sigma_{\theta\theta}\sqrt{\pi l}$ using the value of $\sigma_{\theta\theta}$ at $r = l/2$. Then

$$K_I = \frac{3}{2}\sigma'_{xy}\sqrt{\pi a}\sin\theta\cos\frac{\theta}{2}. \qquad \text{(A2)}$$

The wing cracks will initiate in the direction θ for which K_I is a maximum. Maximising K_I with respect to θ gives $\theta_c = 0.392\,\pi = 70.5°$, so that the stress intensity for the wing crack initiation is

$$K_I = \frac{2}{\sqrt{3}}\sigma'_{xy}\sqrt{\pi a}. \qquad \text{(A3)}$$

The most dangerous crack is that lying at the angle ψ which again maximises K_I. Substituting for σ'_{xy} [equation (2)] and maximising gives $\tan 2\psi = 1/\mu$ from which [using equation (1)] we obtain the maximum value of K_I as

$$K_I = \left\{\frac{\pi a}{3}\right\}^{1/2}\{(\sigma_3 - \sigma_1)(1+\mu)^{1/2} + \mu(\sigma_1 + \sigma_3)\}. \qquad \text{(A4)}$$

Thus, for proportional loading with $\sigma_3 = \lambda\sigma_1$

$$K_I = \frac{-\sigma_1\sqrt{\pi a}}{\sqrt{3}}\{(1-\lambda)(1+\mu^2)^{1/2} - (1+\lambda)\mu\}. \qquad \text{(A5)}$$

This result holds provided σ_{xx} is compressive. When it is tensile, the crack faces separate and the frictional force $\mu\sigma_{xx}$ disappears; but σ_{xx} as well as σ_{xy} is now concentrated by the crack, giving a new term in the equation for the stress intensity. The stress $\sigma_{\theta\theta}$ at a distance r from the crack tip on the plane at angle θ to the crack plane [15] is now

$$\sigma_{\theta\theta} = -\frac{3\sigma_{xy}\sqrt{\pi a}}{2\sqrt{2\pi r}}\sin\theta\cos\theta/2 - \frac{\sigma_{xx}\sqrt{\pi a}}{\sqrt{2\pi r}}\cos^3\theta/2. \qquad \text{(A6)}$$

Defining $K_I = \sigma_{\theta\theta}\sqrt{2\pi r}$ as before gives

$$K_I = -\frac{3}{2}\sqrt{\pi a}\cos\theta/2\{\sigma_{xy}\sin\theta + \sigma_{xx}\cos^2\theta/2\} \qquad \text{(A7)}$$

or

$$K_I = -\frac{3}{2}\sqrt{\pi a}\cos\theta/2\{\tau\sin 2\psi\sin\theta + (\sigma + \tau\cos 2\psi)\cos^2\theta/2\} \qquad \text{(A8)}$$

where $\tau = \frac{1}{2}(\sigma_3 - \sigma_1)$ and $\sigma = \frac{1}{2}(\sigma_3 + \sigma_1)$. Maximising K_I with respect to θ (using equation A7) gives

$$\sigma_{xx}\tan\theta/2 + 2\sigma_{xy}\tan^2\theta/2 - \sigma_{xy} = 0 \qquad \text{(A9)}$$

and then maximising with respect of 2ψ [using equation (A8)] gives

$$\tan 2\psi = 3\tan\theta/2 \qquad \text{(A10)}$$

(discarding the trivial solution $\cos\theta/2 = 0$). This pair of simultaneous equations in θ and ψ defines the critical crack angle and propagation direction.

If only σ_{xx} exists, the only solution of equation (A9) is $\theta = 0$, and, from equation (A10), $\psi = 0$; then the sample fails by a simple tensile fracture. If only σ_{xy} exists (the crack closes, but without friction) then equation (A9) gives $\theta = 70.5°$ and equation (A10) gives $\theta = 45°$ in agreement with the results given earlier. When both σ_{xx} and σ_{xy} exist, a solution is obtained by substituting equation (A10) into (A9), replacing σ_{xx} and σ_{xy} by their definitions, and writing $R = \sigma/\tau\,[= (\lambda+1)/(\lambda-1)]$ to give

$$3R\tan 2\psi - 6\sin 2\psi\tan^2 2\psi = 0. \qquad \text{(A11)}$$

One solution is obvious: it is $\psi = 0$, when, from equations (A10) and (A8) $\theta = 0$ and

$$K_I = \sigma_3\sqrt{\pi a}. \qquad \text{(A12)}$$

This defines the fracture initiation stress in tension. The other solution is found by rewriting equation (A11) in the form

$$\frac{3R}{\cos 2\psi} - 6 + 2\tan^2 2\psi = 0. \qquad \text{(A13)}$$

Replacing $\cos 2\psi$ by $(1 + \tan^2 2\psi)^{-1/2}$ gives a quadratic in $\tan^2 2\psi$

$$4\tan^4 2\psi - (24 + 9R^2)\tan^2 2\psi + 36 - 9R^2 = 0.$$

The solution is

$$\tan^2 2\psi = 3 + \frac{9}{8}R^2 + \frac{3}{8}(64R^2 + 9R^4)^{1/2} \quad \text{where } R < 0$$

$$\tan^2 2\psi = 3 + \frac{9}{8}R^2 + \frac{3}{8}(64R^2 + 9R^4)^{1/2} \quad \text{where } R \geq 0. \qquad \text{(A14)}$$

The sign is selected such that the solution also satisfies equation (A15) (the redundancy arises because of the squaring of R). Substituting equation (A10) into equation (A8) now gives

$$K_I = \frac{\sqrt{\pi a}\,27}{(9 + \tan^2 2\psi)^{3/2}}\{\sigma + \tau(\cos 2\psi + \sin 2\psi\tan 2\psi)\}. \qquad \text{(A15)}$$

Using the definitions of σ and τ gives the equation for the remaining part of the initiation surface

$$\frac{\sigma_1\sqrt{\pi a}}{K_{IC}} = \frac{2(9 + \tan^2 2\psi)^{3/2}}{(\lambda-1)[(\lambda+1)/(\lambda-1) + \cos 2\psi\sin 2\psi\tan 2\psi]}. \qquad \text{(A16)}$$

The fracture initiation surface in stress space is calculated by choosing a value of R (or λ), calculating ψ from (A13), evaluating σ_1 from equation (A16) and then calculating σ_3 from the definition of λ.

Wing crack growth in an infinite plate

Consider the potential energy P of the body containing the kinked crack shown in Fig. 14. The wing cracks are idealised as straight and parallel to the principal axis of compression. Let the crack be pinned together so that no relative motion is possible, and let the sample be loaded to a stress σ_1, σ_3 and then held so that the remote displacements are constant. The elastic energy stored is U_0. Now relax the pinning tractions gradually to zero allowing the faces of the initial crack to slide and faces of the wing cracks to separate and open. As this happens, the tractions do work W. In addition, elastic energy is stored in the body: U_{I} associated with the mode I (wedging) field; and U_{II}, associated with the mode II (sliding) field. The potential energy of the body plus its loads is

$$P = U_0 + U_{\text{I}} + U_{\text{II}} - W. \tag{A17}$$

Before any sliding has taken place, the tractions on the faces of the initial, angled crack, are [see equations (1) and (2)]

$$\begin{aligned} T_s &= \sigma_{xy} + \mu\sigma_{xx} \\ T_n &= \sigma_{xx} \end{aligned} \tag{A18}$$

The tractions acting on the wing cracks are

$$\begin{aligned} T_s &= 0 \\ T_n &= \sigma_3 \end{aligned} \tag{A19}$$

Sliding relaxes T_s over the initial, angled crack, allowing it to do work, but T_n does no work because the displacement is in the plane of the crack. Over the faces of the wing cracks, T_n is relaxed and does work. Since the system is linear-elastic, the total work done is

$$\begin{aligned} W &= \frac{1}{2}\sum \text{Force} \times \text{Displacement} \\ &= \frac{1}{2}\int_{S_1} 2T_s u_s(x)\,\mathrm{d}S_1 + \frac{1}{2}\int_{S_2} 2T_n u_n(x)\,\mathrm{d}S_2 \end{aligned} \tag{A20}$$

where $\pm u_s(x)$ are the shear displacements over the surface S_1 of the angled crack, and $\pm u_n(x)$ are the normal displacements over the surface S_2 of the wing cracks.

Consider the kinked crack to be straightened out to form a straight crack of length $2(l+a)$. The sliding displacements $u_s(x)$ of the faces of the initial crack are

$$u_s(x) = \pm\frac{\alpha_1\delta\{(l+a)^2 - x^2)^{1/2}}{(l+a)} \tag{A21}$$

where x is measured from the centre of the crack, $\alpha_1 (\sim 1)$ is a constant to allow for the kinked shape of the crack and 2δ is the maximum relative sliding displacement. We assume that the normal displacement $u_n(x)$ of each face of the wing crack falls linearly with distance to the tip

$$u_n = \pm\alpha_2\delta\left\{\frac{l+a-x}{l+a}\right\} \tag{A22}$$

(The detailed shape is unimportant: it is taken care of by the second geometric constant α_2.) Inserting this into equation (A20) and integrating gives

$$\begin{aligned} W = \alpha_1(\sigma_{xy} + \mu\sigma_{xx})aB\delta\{[1 - 1/(1+\mathrm{L})^2]^{1/2} \\ + (1+\mathrm{L})\arcsin[1/(1+\mathrm{L}]\} + \alpha_2\sigma_3 aLB\delta \end{aligned}$$

where $L = l/a$. Expanding terms as power series gives

$$\begin{aligned} W = 2\alpha_1(\sigma_{xy} + \mu\sigma_{xx})aB\delta\{1 - 1/6(1+\mathrm{L})^2\} \\ + \alpha_2\sigma_3 aLB\delta. \end{aligned} \tag{A23}$$

The first group of terms is the work done by the shearing tractions T_s acting on the surface of the initial angled crack. This term is almost independent of L: it increases from $\pi\alpha_1/2(\sigma_{xy} + \mu\sigma_{xx})aB\delta$ when $L = 0$ to the limit $2\alpha_1(\sigma_{xy} + \mu\sigma_{xx})aB\delta$ when $L \gg 1$. The second term is the work done against σ_3 as the wing crack opens. The entire equation can be approximated adequately, by

$$W = 2aB\delta\{\alpha_1(\sigma_{xy} + \mu\sigma_{xx}) + \alpha_2\sigma_3 L/2\}. \tag{A24}$$

The displacement δ can be resolved into a component δ_n normal to the "average" plane of the crack (at angle θ to the X_1 axis) and a component δ_s in this plane, where

$$\delta^2 = \delta_n^2 + \delta_s^2 \tag{A25}$$

where, from Fig. 14

$$\delta_n = \delta\sin\theta = \frac{\delta L\sin\psi}{(\sin^2\psi + (\cos\psi + L)^2)^{1/2}} \simeq \frac{L\delta}{\sqrt{2(1+L)}} \tag{A26}$$

$$\delta_s = \delta\cos\theta = \frac{\delta(1 + L\cos\psi)}{[\sin^2\psi + (\cos\psi + L)^2]^{1/2}} \simeq \frac{\delta}{1+L}. \tag{A27}$$

The opening displacement $2\delta_n$, applied to the centre of a crack of length $2(l+a)$ in an infinite body, creates a stress field with an elastic energy, which, on dimensional grounds, must have the form $U_{\text{I}} = \alpha_3 E_0 B\sigma_n^2$ where α_2 is a constant (roughly $4/\pi$) to be determined. Similarly, the shear displacement $2\delta_s$ at the centre of the crack creates an additional field, with an elastic energy of $U_{\text{II}} = \alpha_3 E_0 B\delta_s^2$. The total elastic energy U_T is simply the sum of the energies of each field taken alone, because the displacements which caused them are orthogonal. Thus

$$U_T = \alpha_3 E_0 B\delta^2. \tag{A28}$$

The extent of sliding is obtained by finding the minimum in the potential energy of the body with respect to δ (holding the crack length constant), using

$$\frac{\mathrm{d}}{\mathrm{d}\delta}\{U_0 + U_T - W\} = 0.$$

Inserting equation (A24) for W and equation (A28) for U_T, and neglecting U_0 (because it is independent of δ at constant crack length) gives

$$\delta = \frac{a}{E_0}\left\{\frac{\alpha_1}{\alpha_3}(\sigma_{xy} + \mu\sigma_{xx}) + \frac{\alpha_2}{2\alpha_3}\sigma_3 L\right\}. \tag{A29}$$

In simple compression ($\sigma_3 = 0$) the displacement is

$$\delta = -\frac{a\sigma_1}{2E_0}\frac{\alpha_1}{\alpha_3}$$

which is independent of the length of the wing cracks. This is a result of the approximation of equation (A24) for equation (A23), and is justified because the dependence on L in equation (A23) is a very weak one. Equation (A29) gives the limiting sliding displacement δ as L becomes large compared to unity.

The sliding displacement wedges the wing cracks open. K_{I}^n, the normal displacement, δ_n, is considered to be applied at the centre of a crack of length $2(l+a)$ so that (18)

$$K_{\text{I}}^n = \frac{\beta_n E_0 \delta_n}{(l+a)^{1/2}} \tag{A30}$$

where $\beta_n \simeq 0.4$. The shearing part of the displacement, δ_s, also generates a mode 1 stress intensity at the tip of a small wing crack [18] of

$$K_{\text{I}}^s = \frac{\beta_s E_0 \delta_s}{(l+a)^{1/2}} \tag{A31}$$

where $\beta_s \simeq 1.0$. The observations of crack propagation shown in Section 2 show that the crack initiates at an angle 70° to the starter crack, in response to the stress intensity K_{I}^s [equation (A31)]. But after the crack has extended by

about a, it rotates into a direction parallel to the axis of principal compression; from there on, its growth is dominated by the stress intensity K_I^n generated by the wedging action [equation (A30)]. This change is allowed for by multiplying K_I^s by the factor $1/(1+L)^{1/2}$ so that when $l=a$, K_I^s is reduced by the factor $1/\sqrt{2}$. (This assumption only affects the result when L is small and a tolerable fit is obtained without it.) The total stress intensity at the crack tip is then

$$K_I = K_I^n + K_I^s/(1+L)^{1/2}. \quad \text{(A32)}$$

Using the equations for K_I^n and K_I^s together with that for δ [equation (A29)] and the definitions of δ_n and δ_s [equations (A26) and (A27)] gives

$$K_I = \frac{\sqrt{\pi a}}{\sqrt{\pi}(1+L)^{3/2}}\left\{\frac{\alpha_1}{\alpha_3}(\sigma_{xy}+\mu\sigma_{xx})+\frac{\alpha_2}{2\alpha_3}\sigma_3 L\right\} \times\left\{\beta_n L+\frac{1}{(1+L)^{1/2}}\right\}. \quad \text{(A33)}$$

The two limits of small and large L are now used to determine the constants α_1/α_3 and α_2/α_3. When $L=0$, equation (A33) reduces identically to the initiation criterion [equation (A3)], provided

$$\frac{\alpha_1}{\alpha_3}=\frac{2\sqrt{2\pi}}{\sqrt{3}}. \quad \text{(A34)}$$

When $L \gg 1$ and σ_3 is positive (i.e. tensile), the stress intensity K_I must reduce to $\sigma_3\sqrt{\pi l}$. This it does provided

$$\frac{\alpha_2}{\alpha_3}=\frac{2\sqrt{2\pi}}{\beta_n}. \quad \text{(A35)}$$

Assembling these results gives the stress intensity for wing crack growth

$$K_I = \frac{\sqrt{\pi a}}{(1+L)^{3/2}}\left\{\frac{2}{\sqrt{3}}(\sigma_{xy}+\mu\sigma_{xx})+\frac{\sigma_3 L}{\beta_n}\right\} \times\left\{\beta_n L+\frac{1}{(1+L)^{1/2}}\right\} \quad \text{(A36)}$$

where $\beta_n = 0.4$ [18]. Replacing σ_{xx} and σ_{xy} by their definitions [equation (1)] and writing $\lambda=\sigma_3/\sigma_1$ gives equation (6) of the main text.

Crack growth from an angled crack in a finite plate: interaction with surfaces

Consider a plate of height h, width w and depth b, containing a central crack of length $2a$, and loaded with principal stresses σ_1 and σ_3, as shown in Fig. 17. Wing cracks of length l extend from the initial crack. If the dimensions of the plate are much larger than the length of the crack, the stress intensity at its tip is given by equation (A36). But if the sample is narrow the crack splits it into vertical beams or columns of length $2(l+\alpha a)$ where $\alpha=\cos\psi=1/\sqrt{2}$ when the initial crack lies at 45° to the principal axes. When loaded, the beams bend outward as shown in the photograph of Fig. 9, and schematically in Fig. 17.

The outward bending allows a greater sliding displacement at the initial crack, and increases the stress intensity at the tips of the wing cracks. We calculate this increase approximately by using standard results of beam theory. One beam is idealised in Fig. 17(b). It consists of two half-beams, each of length $(l+\alpha a)$, joined end-to-end. The upper beam has average width $\frac{1}{2}(w+\alpha a)$; the lower beam, $\frac{1}{2}(w-\alpha a)$. A sliding displacement δ causes outward and downward displacements of the centre of the beam; the outward displacement bends the beam, the downward displacement extends the upper half beam and compresses the lower half beam slightly, storing elastic energy.

We calculate the extent of sliding by minimising the elastic energy of the system with respect to the sliding displacement. Strictly, we consider an initial crack at an arbitrary angle ψ, and minimise the elastic energy first with respect to ψ at fixed δ (giving the angle of the most damaging crack) and then minimising with respect δ to give the actual displacement. But, for simplicity, we note that experiments (Fig. 4) show that an initial crack at an angle ψ of 45° is the most damaging when the crack is long. Then the outward and downward displacements, δ_n, are equal as shown in Fig. 17(a).

Consider, now, the sample of Fig. 17(a), with the crack welded shut. The sample is loaded in simple compression to the stress σ_1, and then the loaded surfaces are held fixed. The crack surfaces are now released so that a sliding displacement 2δ takes place. The beams bend outward so that the mid-point displacement of one beam is δ_n, and the compressional elastic energy in the upper half beam decreases, and that in the lower half beam increases, because of the downward displacement δ_n. The change in elastic energy of one beam (as in Fig. 6) is easily shown to be

$$U=\frac{bt^3\delta_n^2 E}{(l+\alpha a)^3}+\sigma b t\delta_n\left\{\frac{\delta_n E}{\sigma_1(l+\alpha a)}-\frac{\alpha a}{t}\right\}. \quad \text{(A37)}$$

Minimising with respect to δ_n gives

$$\delta_n=\frac{\sigma_1}{2E}\frac{\alpha a}{t}\frac{(l+\alpha a)}{[1+t^2/(l+\alpha a)^2]}. \quad \text{(A38)}$$

Replacing this in equation (A37) and differentiating with respect to l (and assuming $t^2/l^2 \ll 1$), gives the crack driving force

$$G=-\frac{\delta U}{b\delta l}=\frac{1}{4}\frac{\sigma_1^2}{E}\frac{(\alpha a)^2}{t} \quad \text{(A39)}$$

and the associated stress intensity is

$$K_I^{B1}=\sqrt{EG}=-\frac{1}{2\sqrt{2\pi}}\left(\frac{a}{t}\right)^{1/2}\sigma_1\sqrt{\pi a} \quad \text{(A40)}$$

where we have set $\alpha=1/\sqrt{2}$ and have taken the negative root to maintain consistency with the definition σ_1 as positive when tensile.

A confining stress, σ_3, applies a distributed load ($\sigma_3 b$ per unit length) to the outer surface of the beam of Fig. 17(b). This generates additional forces, tending to limit the sliding displacement and reduce the stress intensity K_I. Its contribution can be calculated by standard methods of beam-theory. This is done in Ref. [10] [equation (A13) to (A14) with the result that a confining stress σ_3 reduces K_I by

$$K_I^{B3}=+\left(\frac{4}{3\pi}\right)^{1/2}\left(\frac{\alpha a}{t}\right)^{3/2}(L+\alpha)^2\sigma_3\sqrt{\pi a}. \quad \text{(A41)}$$

The stress intensity due to interaction with the surface is the sum of equation (A40) and (A41), with $\alpha=1/\sqrt{2}$. We approximate the total stress intensity by the larger of this, and of equation (A41).

There is one additional factor. We have assumed, until now, that the outward deflection of the beams was not enough to change their geometry, and thus the moments acting on them. But when the cracks are long and the loads high, large deflections increase the bending moments exerted by σ_1. At sufficiently high loads the beams buckle and deflections (and K_I) become very large indeed; the Euler buckling load for an end-loaded beam built-in at both ends (as in Fig. 17) is

$$P_{crit}=\frac{\pi^2 EI}{(l+\alpha a)^2}.$$

A prefectly straight, uniform, beam buckles suddenly and catastrophically at P_{crit}; but one which is already bent collapses progressively so that the deflections become increasingly large as the end load P approaches P_{crit}. We

linearise this progressive collapse (it is the first-order solution to the problem treated in this section, allowing for the change of geometry of the beam as it bends) by multiplying the bending deflection δ and K_I^B (which depends linearly on δ) by the correction factor

$$f = \frac{1}{1 - \dfrac{P}{P_{crit}}} = \frac{1}{\left(1 + \dfrac{12}{\pi^2}\dfrac{(l+\alpha a)^2}{t^2}\dfrac{\sigma_1}{E}\right)}. \quad \text{(A42)}$$

Equation (8) of the text follows immediately.